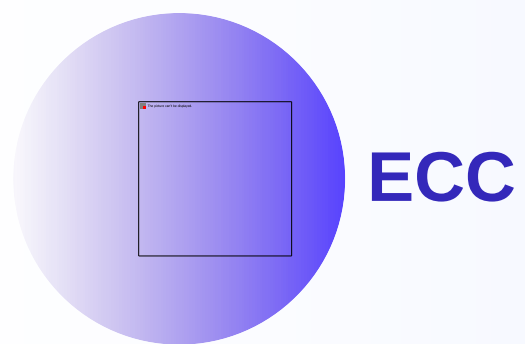


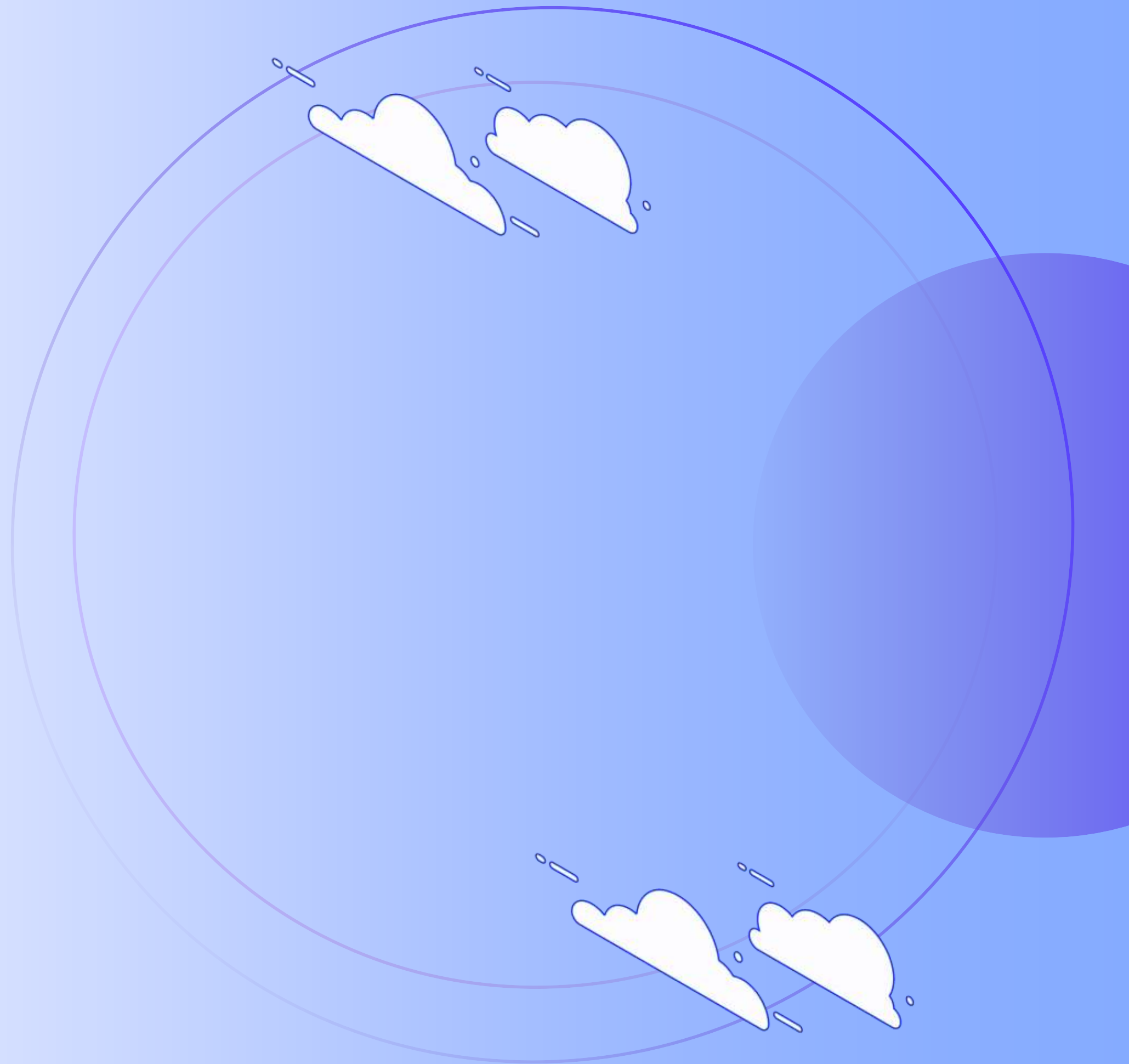


ELLIPTIC

CURVE

CRYPTOGRAPHY

PHY

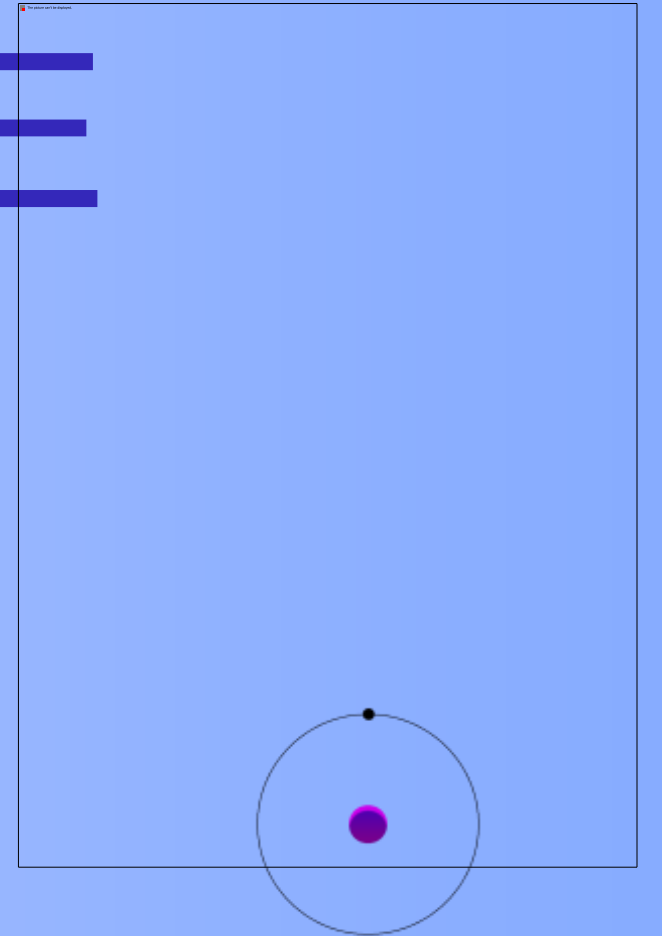
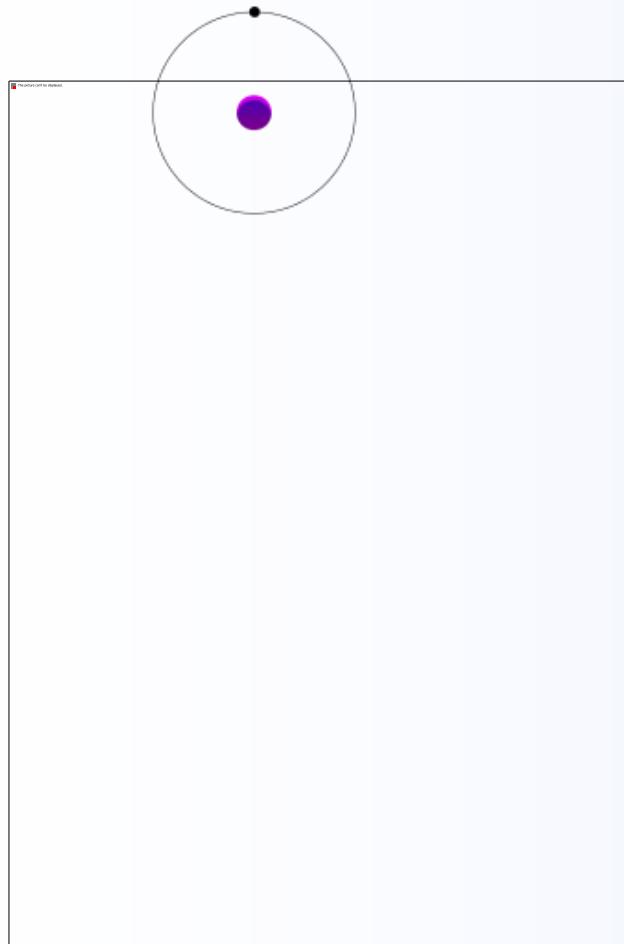


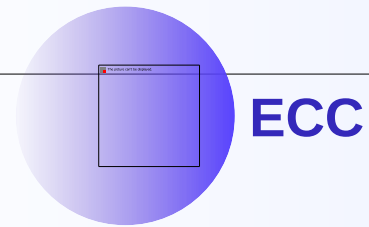


ECC: SECURITY MADE EFFICIENT

The power of ECC lies in these core advantages:

- The internet's rise necessitates strong cryptography.
 - Traditional methods often rely on shared private keys.
 - Securely sharing these keys has been a long-standing problem.
- ECC offers a powerful and efficient alternative.





ELLIP

TIC CURVE

S

- Elliptic curves emerged in the 18th century from investigations into elliptic integrals (related to ellipse perimeters).
- Elliptic integrals led to the development of elliptic functions and the theory of elliptic curves.
- Elliptic curves are historically, not mathematically, related to ellipses.
- Elliptic curves resemble cubic functions, not quadratic functions like ellipses.



WEIERSTR EQUATION

- Any elliptic curve follows the Weierstrass Equation:

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

- For fields without a characteristic of 2 or 3, this simplifies to:

$$y^2 = x^3 + Ax + B$$



GROUPS



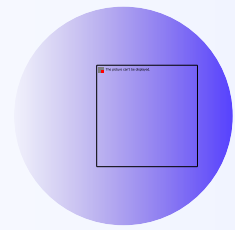
A group is a set of elements that follows the following axioms:

- Closure: For all elements a and b in the group, $a \cdot b$ is also in the group.
- Associativity: For all elements a , b , and c in the group, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- Identity Element: There exists an element e in the group such that for any element a ,
$$a \cdot e = e \cdot a = a.$$
- Inverse Element: For each element a in the group, there exists an element a^{-1} such that
$$a \cdot a^{-1} = a^{-1} \cdot a = e.$$

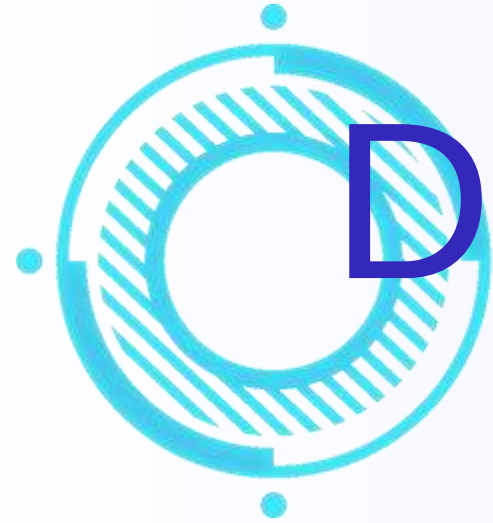


ELLIPTIC CURVE GROUP

- Point Addition : Defined geometrically, not arithmetically.
- $P + Q$ ($P \neq Q$): Line through P , Q intersects curve at R' ; R is reflection of R' over x-axis.
- $P + P$: Tangent at P intersects curve at R' ; R is reflection of R' over x-axis.
 - Identity : "Point at infinity" (O): $P + O = O + P = P$.
 - Inverse : $-P$ is reflection of P over x-axis: $P + (-P) = O$.
- Scalar multiplication in EC group is repeated point addition ($kP = P + P + \dots + P$, k times), easy to compute but hard to reverse, forming the basis of ECC security.

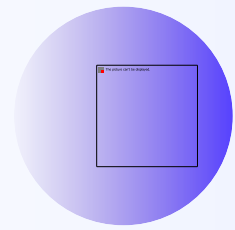


ECC



DIFFIE HELLMAN

- A way to share private keys.
- Normally uses exponential arithmetic.
- If ECC is used instead, it will provide smaller key sizes, and better security.
- Each participant will have a private number a , b respectively.
- Each participant will calculate a point A , B respectively as $A = aG$, $B = bG$ [G is the base point].
- A , B are sent in a public channel knowing that no one will be able to generate a or b from it.



ECC



APPLICATIONS

- 1-ECDSA Crypto System
- 2- ECDH Crypto System
- 3-Bitcoin

1-ECDSA Crypto

System:



What is ECDSA?

- A digital signature algorithm based on Elliptic Curve Cryptography (ECC).
 - Provides authentication, integrity, and non-repudiation.
 - Used in Bitcoin, Ethereum, TLS, and secure messaging apps.

Why Use ECDSA?

- High security with smaller key sizes (e.g., 256-bit ECC $\approx$ 3072-bit RSA).
 - Efficient in computation and bandwidth.
 - Widely adopted in modern cryptographic protocols.

⚙️ How Does ECDSA

Work?

Key Generation:

- Private key: Random number d .
- Public key: $Q = d \times G$ (point multiplication on the curve).

Signing:

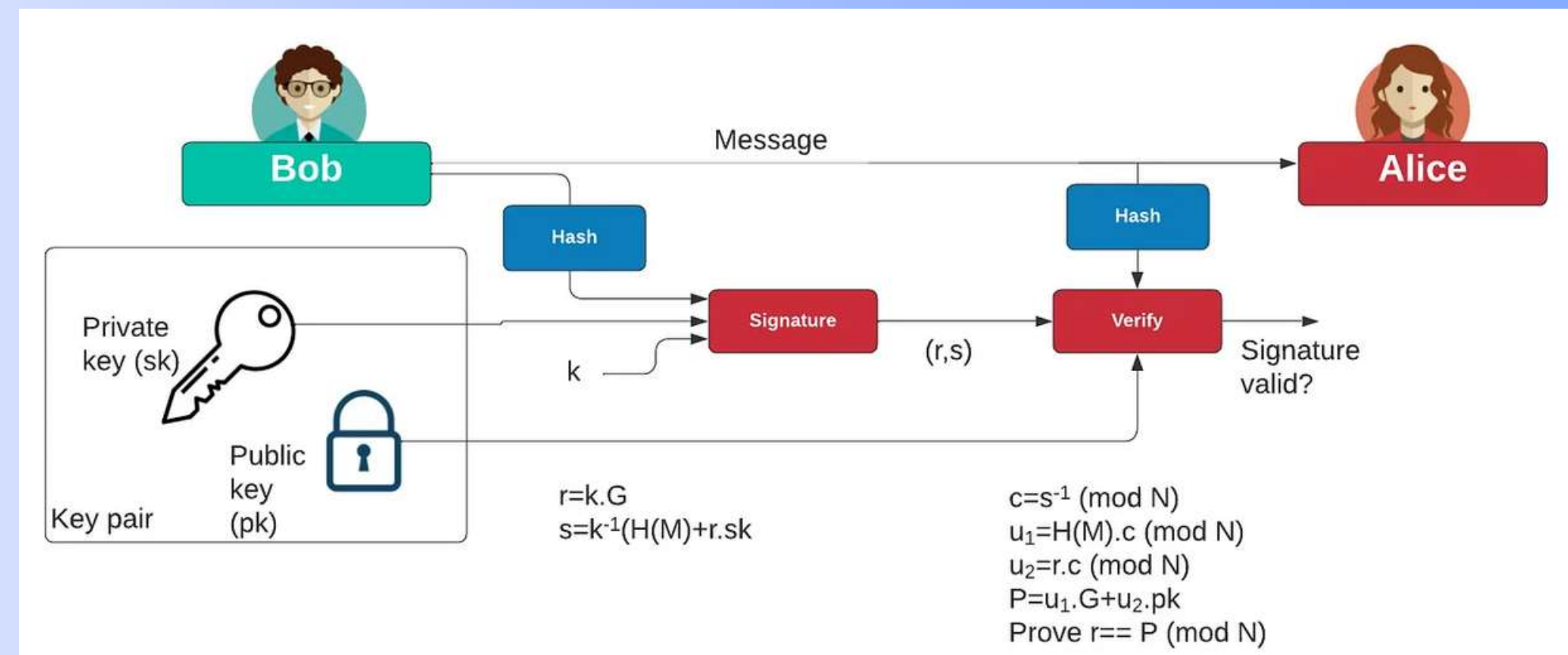
- Uses the private key to create a unique signature (r, s) for each message.

Verification:

- Anyone with the public key can verify the signature.

Uses

- Cryptocurrencies (e.g., Bitcoin, Ethereum)
 - Secure communications (TLS/SSL)
 - Digital certificates
 - Mobile & IoT security



ECDH Crypto System:

What is ECDH?

- A key exchange protocol based on Elliptic Curve Cryptography (ECC).
- Allows two parties to securely share a secret key over an insecure channel.
- Does not provide authentication by itself—usually combined with other methods.

Why Use ECDH?

- Stronger security with smaller keys (e.g., 256-bit ECC $\approx$ 3072-bit RSA).
 - Faster and more efficient than traditional Diffie-Hellman.
- Ideal for mobile devices and resource-constrained environments

⚙️ How Does ECDH Work?

Each party generates:

- A private key (random number)
- A public key (using $Q = d \times G$)

Key Exchange:

- Each party shares their public key.

Shared Secret:

- Each computes the secret using their private key and the other's public key.
- Both arrive at the same shared key due to ECC properties.

Q uses

- Secure messaging apps (e.g., Signal, WhatsApp)
 - TLS (HTTPS)
 - VPNs
- Encrypted emails and file sharing

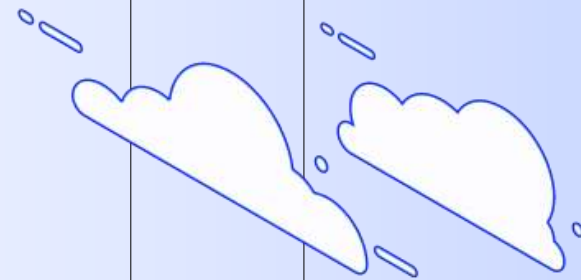


This is the difference between ECDSA and ECDH

Feature		ECDSA	ECDH
 Purpose		Digital Signatures (Authentication)	Key Exchange (Confidentiality)
 Function		Signs and verifies messages	Derives shared secret over insecure channels
 Uses Private Key		To sign messages	To compute shared secret
 Key Concept		Message integrity and authenticity	Secure key agreement between two parties



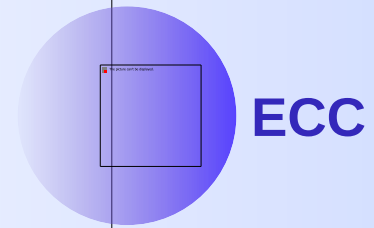
BITCOIN APPLICATIONS





BITCOIN

ECC is used for a variant to derive public keys and sign Bitcoin transactions. Bitcoin uses the Elliptic Curve, “secp256k1”. $y^2 = x^3 + 7$. The Public Key is easy to derived but impossible for robustness of the system. It comes from the Private Key and a constant G known as the Generator Point : K (Public Key) = k (Private Key) $\times$ G (Generator Point). In this particular case, multiplying is best interpreted as adding the Generator Point as many times as indicated by the private key value. For example, If the Private Key were «5», to obtain the Public Key, you would have to add $G + G + G + G + G = 5G$ that makes it a constant already determined in the initial parameters of the secp256k1 elliptic curve:



THANK

YOU!